Unidade 2 - BBox

Aplicação { RN - mundo
Grafos { = BBox Tipo { EXTERNA / INTERNA

Boundaring Box

Utilidade: ? problema
? Grandeza

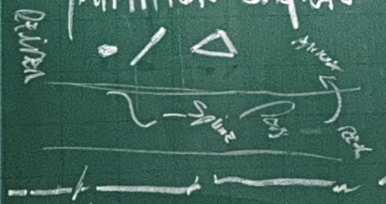
Proc. mio

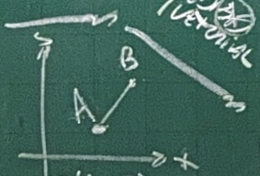
$r^2 = x^2 + y^2$

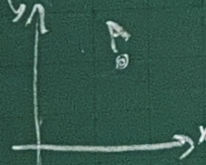
? Cenas — "REAL" / "Aproximada"

$f(x)$

? primitivas gráficas





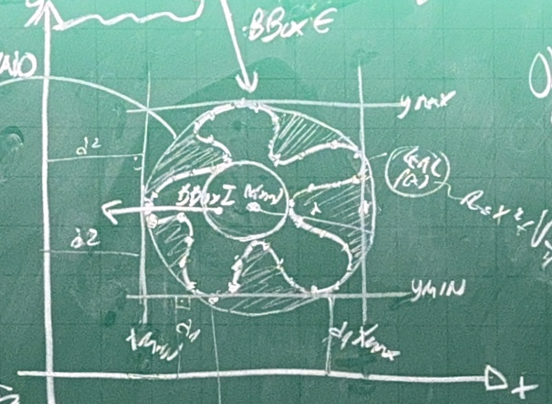


A(20,15)

Xmin = 20
Xmax = 20
Ymin = 15
Ymax = 15

A(10,5)
B(30,25)

Xmin = 10
Xmax = 30
Ymin = 5
Ymax = 25



BBox E

Sen { 2D (A) / Vetorial

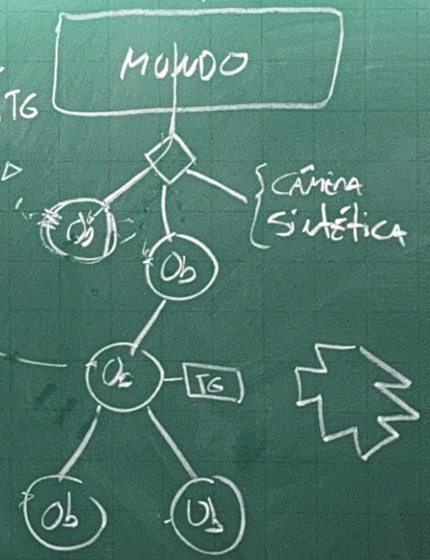
BBox { Xmin, Xmax, Ymin, Ymax, Zmin, Zmax

"Lados" marcados pelos eixos

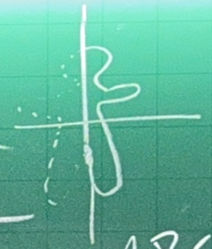
Custo { Custando BBox / Cálculo / Aplicação

Obj. Gráfico
Atributo { BBox / mundo.TG / Matriz

Dados
(Cenas) ? Grafo de Cena



Unária	Shift
+	−
*	/
√	^



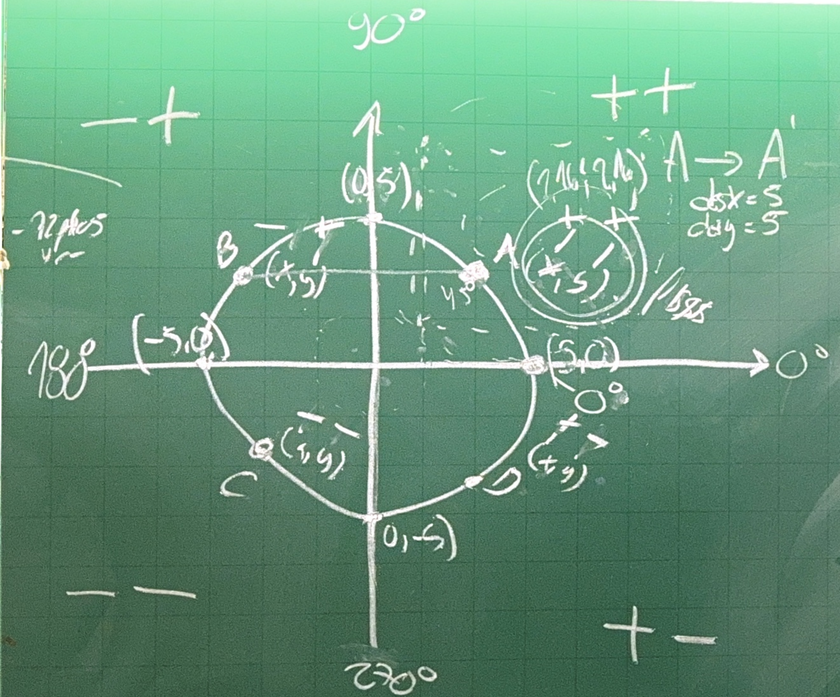
A, B, C, D

Exercício 1

4 pts ? 8x calcular seno cos pts (A)

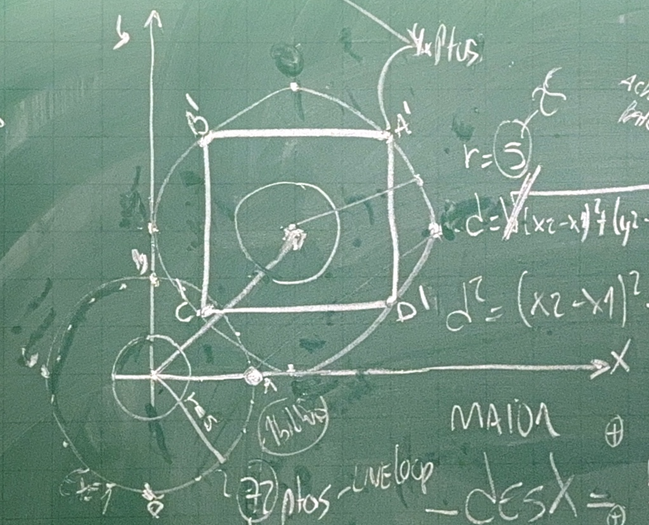
1 pto 2x Unária

1x x = y



(A) Desenhar

- Círculo maior ⇒ raio = 5
- Círculo menor ⇒ raio = 2
- Quadrado



(b) Testar seu círculo

X Muitas vezes

(B1) BBOX (aproximação)

xmin:
xmax:
ymin:
ymax:

IF (Mouse x > xMIN) AND
(Mouse x < xMAX) AND
(Mouse y > yMIN) AND
(Mouse y < yMAX) → Vend dentro

FALSO

Acha raio

(B2) Círculo MA (REAL)

$r = 5$

$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

$d^2 = (x_2-x_1)^2 + (y_2-y_1)^2$

MAIOR

desx = 5
desy = 5

MENOR

desx = 5
desy = 5